Basics of FE-Analysis BK10A6400 Lecture 9: Q8 plane element and its numerical integration over volume (area)

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General things

- Status of bonus tasks:
  - Bonus task 4 (beam structure): 50/141 returned
  - Bonus task 5 (plane element): deadline today, around 40/140 returned
  - Bonus task 6 (plane element with selective integration)
- Lectures left:
  - 24.11: Solid elements etc.
  - 1.12: Linearized dynamics
  - 8.12: (Full course revision, tips for the exam)

Today

- Biquadratic plane element (Q8)
- Equivalent loadings (Bonus task?)

Quadratic plane element Q8

Polynomial terms form Pascal's triangle:

$$(x+y)^n$$

$n=0$:					1				
$n=1$:				ξ		η			
$n=2$:			ξ^2		$\xi\eta$		η^2		
$n=3$:		ξ^3		$\xi^2\eta$		$\xi\eta^2$		η^3	
$n=4$:	ξ^4		$\xi^3\eta$		$\xi^2\eta^2$		$\xi\eta^3$		η^4

Displacement field (polynoms form Pascal's triangle):

$$\begin{cases} u_h(\xi,\eta) := a_0 + a_1\xi + a_2\eta + a_3\xi\eta + a_4\xi^2 + a_5\eta^2 + a_6\xi^2\eta + a_7\xi\eta^2 \\ v_h(\xi,\eta) := b_0 + b_1\xi + b_2\eta + b_3\xi\eta + b_4\xi^2 + b_5\eta^2 + b_6\xi^2\eta + b_7\xi\eta^2 \end{cases}.$$

Displacement interpolation is the same for both displacements $a \Rightarrow$ same shape functions. Eight unknowns, eight eqs:

$$\begin{cases} u_h(-1,-1) := a_0 - a_1 - a_2 + a_3 + a_4 + a_5 - a_6 - a_7 = u_1 \\ u_h(1,-1) := a_0 + a_1 - a_2 - a_3 + a_4 + a_5 - a_6 + a_7 = u_2 \\ u_h(1,1) := a_0 + a_1 + a_2 + a_3 + a_4 + a_5 + a_6 + a_7 = u_3 \\ u_h(-1,1) := a_0 - a_1 + a_2 - a_3 + a_4 + a_5 + a_6 - a_7 = u_4 \\ u_h(0,-1) := a_0 - a_2 + a_5 = u_5 \\ u_h(1,0) := a_0 + a_1 + a_4 = u_6 \\ u_h(0,1) := a_0 + a_2 + a_5 = u_7 \\ u_h(-1,0) := a_0 - a_1 + a_4 = u_8 \end{cases}.$$

Q8

By solving a_i:s:

$$\boldsymbol{a} = \boldsymbol{A}^{-1}\boldsymbol{u}_u \tag{1}$$

and substituting them into the Eq.(30):

$$u_h = \boldsymbol{a}^T\boldsymbol{p} = \boldsymbol{A}^{-T}\boldsymbol{u}_u^T\boldsymbol{p}^T = \underbrace{\boldsymbol{A}^{-T}\boldsymbol{p}}_{\boldsymbol{N}}\boldsymbol{u}_u \tag{2}$$

where $\boldsymbol{N}$ includes shape functions.

$$\begin{aligned} W_{int} &= \frac{1}{2}\boldsymbol{u}^T \underbrace{l_z \int_x \int_y \boldsymbol{B}^T \boldsymbol{D} \boldsymbol{B}\, \mathrm{d}y \mathrm{d}x}_{\overline{\boldsymbol{K}}}\, \boldsymbol{u} \\ &= \frac{1}{2}\boldsymbol{u}^T \underbrace{l_z \int_{-1}^{1} \int_{-1}^{1} \boldsymbol{B}^T \boldsymbol{D} \boldsymbol{B}\; \det\left(\boldsymbol{J}_e\right) \mathrm{d}\xi \mathrm{d}\eta}_{\overline{\boldsymbol{K}}}\, \boldsymbol{u} \end{aligned} \tag{3}$$

or more generally, as (and no need to make integration at the component level):

$$W_{int} = \frac{1}{2} \int_V \boldsymbol{\sigma} \boldsymbol{\varepsilon}\, \mathrm{d}V = \frac{1}{2} l_z \int_{-1}^{1} \int_{-1}^{1} \boldsymbol{\sigma} \boldsymbol{\varepsilon} \det\left(\boldsymbol{J}_e\right) \mathrm{d}\xi \mathrm{d}\eta \tag{4}$$

$$\boldsymbol{f}_{int} = \frac{\partial W_{int}}{\partial \boldsymbol{u}} \quad ; \quad \overline{\boldsymbol{K}} = \frac{\partial \boldsymbol{f}_{int}}{\partial \boldsymbol{u}}$$

Q8: Numerical integration

How many Gaussian points are needed?
Interpolation: max polynomial order $\rightarrow 2$, $\boldsymbol{\varepsilon}^T \boldsymbol{\varepsilon} \rightarrow 4$
The degree of a polynomial of the integrand $p = 4$.

$$nip = \frac{(p+1)}{2} = 2.5 \tag{5}$$

so three integration points are needed.

Distributed forces

- Distributed equally on nodes
- Distributed by using shape functions

The variation (i.e virtual work) for the externally applied forces:

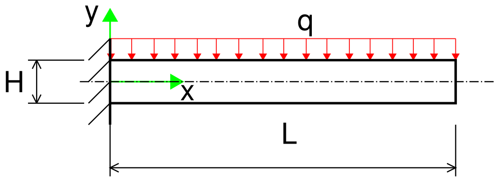
$$\delta W_{ext} = \boldsymbol{f}_{ext} \delta \boldsymbol{u}_h = \int_V \boldsymbol{b}^T \boldsymbol{N}_m \, \mathrm{d}V \, \delta \boldsymbol{u} \tag{6}$$

where $\boldsymbol{b}$ is a vector of body forces $[N/m^3]$.

$$\boldsymbol{f}_{ext} = \int_V \boldsymbol{b}^T \boldsymbol{N}_m \, \mathrm{d}V \tag{7}$$

How do we distribute forces from continuous loading (structure's own mass, snow load, pressure, contact forces etc) on nodal DOFs?

- with "lumped" approach
- with shape functions



Principle of Virtual Work (Variational form)

$$\delta W_{int} = \delta W_{ext} \tag{8}$$

Internal virtual work:

$$\delta W_{int} = \int_V \boldsymbol{\sigma}^T \delta \boldsymbol{\varepsilon} \tag{9}$$

External virtual work:

$$\delta W_{ext} = \int_V \boldsymbol{b}^T \delta \boldsymbol{u}_h \, \mathrm{d}V + \int_\Gamma \boldsymbol{t}^T \delta \boldsymbol{u}_h \, \mathrm{d}\Gamma + \sum \boldsymbol{f}_i \delta \boldsymbol{u}_i \tag{10}$$

where $\boldsymbol{b}$ is a vector of body forces (N/m^3), $\boldsymbol{t}$ is a vector of traction forces (surface forces (N/m^2)), $\boldsymbol{f}_i$ is point forces (N) and $d\Gamma$ is the differential size of the surface or line under traction forces. Virtual displacement (field) can be approximated as

$$\delta \boldsymbol{u}_h = \boldsymbol{N}_m \delta \boldsymbol{u}$$

Example: distributed load q

Let's calculate equivalent nodal forces from a constant line load $q[N/m]$. A vector of equivalent nodal forces can be computed as:

$$\delta W_{ext}^{qf} = \int_{\Gamma} \boldsymbol{t}^T \delta \boldsymbol{u}_h \, \mathrm{d}\Gamma = \underbrace{\int_{x} \boldsymbol{q}^T \boldsymbol{N}_m \, \mathrm{d}x}_{\boldsymbol{f}_{equiv}^{qf}} \, \delta \boldsymbol{u} \tag{11}$$

where $\boldsymbol{q} = [0, q]^T$. Equivalent nodal forces can be written as

$$\boldsymbol{f}_{equiv}^{qf} = \int_{-1}^{1} \boldsymbol{q}^T \boldsymbol{N}_m J_{e_{11}} \, \mathrm{d}\xi \tag{12}$$

where $J_{e_{11}}$ describes the transformation between reference configuration (binormalized coordinates) and initial configuration (physical coordinates) and can also be taken from $\boldsymbol{J}_e$.

Boundary (segment) Jacobian for the element boundary

Isoparametric transformation (mapping) between bi-normalized (natural) coordinates $\boldsymbol{\xi} = \{\xi, \eta\}^T$ and the initial (physical) configuration:

$$\boldsymbol{X}_h(\boldsymbol{\xi}) = \begin{pmatrix} X_h(\xi,\eta) \\ Y_h(\xi,\eta) \end{pmatrix}, \qquad \boldsymbol{J}_e = \frac{\partial \boldsymbol{X}_h}{\partial \boldsymbol{\xi}} = \begin{pmatrix} X_{h,\xi} & X_{h,\eta} \\ Y_{h,\xi} & Y_{h,\eta} \end{pmatrix}.$$

Consider now an element boundary segment parametrised by a single natural coordinate $s \in [-1, 1]$:

$$\boldsymbol{\xi}(s) = \begin{pmatrix} \xi(s) \\ \eta(s) \end{pmatrix}, \qquad \boldsymbol{X}_h(s) = \boldsymbol{X}_h(\boldsymbol{\xi}(s)).$$

Boundary (segment) Jacobian for the element boundary

The tangent vector on the boundary segment follows from the chain rule:

$$\frac{\mathrm{d}\boldsymbol{X}_h}{\mathrm{d}s} = \frac{\partial \boldsymbol{X}_h}{\partial \boldsymbol{\xi}} \frac{\mathrm{d}\boldsymbol{\xi}}{\mathrm{d}s} = \boldsymbol{J}_e\,\boldsymbol{J}_{bnd}, \qquad \boldsymbol{J}_{bnd} = \frac{\mathrm{d}\boldsymbol{\xi}}{\mathrm{d}s} = \begin{pmatrix} \frac{\mathrm{d}\xi}{\mathrm{d}s} \\ \frac{\mathrm{d}\eta}{\mathrm{d}s} \end{pmatrix}.$$

The physical line element on the boundary is

$$\mathrm{d}\Gamma = \left\| \frac{\mathrm{d}\boldsymbol{X}_h}{\mathrm{d}s} \right\| \mathrm{d}s = J_{\text{bnd}}(s)\,\mathrm{d}s,$$

where the boundary scaling factor between is

$$\boxed{J_{\text{bnd}}(s) = \|\boldsymbol{J}_e\,\boldsymbol{J}_{bnd}\| = \sqrt{\left(\frac{\mathrm{d}X_h}{\mathrm{d}s}\right)^2 + \left(\frac{\mathrm{d}Y_h}{\mathrm{d}s}\right)^2}.}$$

Boundary (segment) Jacobian: Q4 examples

For a Q4 element:

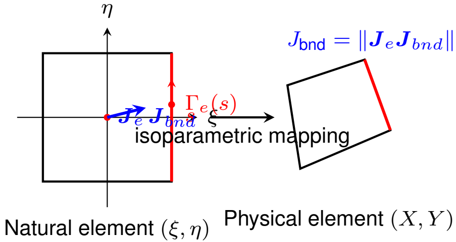
- **Top boundary segment:** $\eta = 1$, $s = \xi$

$$\boldsymbol{J}_{bnd} = \{1, 0\}^T, \qquad J_{\mathsf{bnd}} = \sqrt{X_{h,\xi}^2 + Y_{h,\xi}^2}.$$

- **Right boundary segment:** $\xi = 1$, $s = \eta$

$$\boldsymbol{J}_{bnd} = \{0, 1\}^T, \qquad J_{\mathsf{bnd}} = \sqrt{X_{h,\eta}^2 + Y_{h,\eta}^2}.$$

Isoparametric mapping: natural binormalized coordinates $\rightarrow$ physical coordinates



Boundary (segment) Jacobian for the element boundary Γ

Isoparametric transformation (mapping) between bi-normalized (natural) coordinates $\boldsymbol{\xi} = \{\xi, \eta\}^T$ and the initial (physical) configuration:

$$\boldsymbol{X}_h(\boldsymbol{\xi}) = \begin{pmatrix} X_h(\xi,\eta) \\ Y_h(\xi,\eta) \end{pmatrix}, \qquad \boldsymbol{J}_e = \frac{\partial \boldsymbol{X}_h}{\partial \boldsymbol{\xi}} = \begin{pmatrix} \dfrac{\partial X_h}{\partial \xi} & \dfrac{\partial X_h}{\partial \eta} \\ \dfrac{\partial Y_h}{\partial \xi} & \dfrac{\partial Y_h}{\partial \eta} \end{pmatrix}. \tag{13}$$

Consider now an element edge parametrised by a single natural coordinate $s \in [-1, 1]$:

$$\boldsymbol{\xi}(s) = \begin{pmatrix} \xi(s) \\ \eta(s) \end{pmatrix}, \qquad \boldsymbol{X}_h(s) = \boldsymbol{X}_h(\boldsymbol{\xi}(s)). \tag{14}$$

Boundary (segment) Jacobian for the element boundary Γ

The tangent vector on the edge follows from the chain rule:

$$\frac{\mathrm{d}\boldsymbol{X}_h}{\mathrm{d}s} = \frac{\partial \boldsymbol{X}_h}{\partial \boldsymbol{\xi}} \frac{\mathrm{d}\boldsymbol{\xi}}{\mathrm{d}s} = \boldsymbol{J}_e \, \boldsymbol{J}_{bnd}, \qquad \boldsymbol{J}_{bnd} = \frac{\mathrm{d}\boldsymbol{\xi}}{\mathrm{d}s} = \begin{pmatrix} \frac{\mathrm{d}\xi}{\mathrm{d}s} \\ \frac{\mathrm{d}\eta}{\mathrm{d}s} \end{pmatrix}. \tag{15}$$

The physical line element on the edge is

$$\mathrm{d}\Gamma = \left\| \frac{\mathrm{d}\boldsymbol{X}_h}{\mathrm{d}s} \right\| \mathrm{d}s = J_{\text{edge}}(s)\, \mathrm{d}s, \tag{16}$$

where the *boundary Jacobian* is

$$\boxed{J_{\text{edge}}(s) = \|\boldsymbol{J}_e \, \boldsymbol{c}\| = \sqrt{\left(\frac{\mathrm{d}X_h}{\mathrm{d}s}\right)^2 + \left(\frac{\mathrm{d}Y_h}{\mathrm{d}s}\right)^2}.} \tag{17}$$

Boundary (segment) Jacobian for the element boundary Γ

For example, for a Q4 element:

- Top edge: $\eta = 1$, $s = \xi \Rightarrow \boldsymbol{c} = \{1, 0\}^T$, so $J_{\text{edge}} = \sqrt{(X_{h,\xi})^2 + (Y_{h,\xi})^2}$.
- Right edge: $\xi = 1$, $s = \eta \Rightarrow \boldsymbol{c} = \{0, 1\}^T$, so $J_{\text{edge}} = \sqrt{(X_{h,\eta})^2 + (Y_{h,\eta})^2}$.

Example: structure's own mass as loading

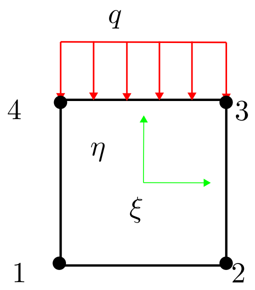
Let's calculate equivalent nodal forces from gravity g [m/s^2]. Then body forces $\boldsymbol{b} = \rho \boldsymbol{g}$ where ρ is density [kg/m^3] and $\boldsymbol{g}$ is a field of gravity [m/s^2]. A vector of equivalent nodal forces due to gravity can be computed as:

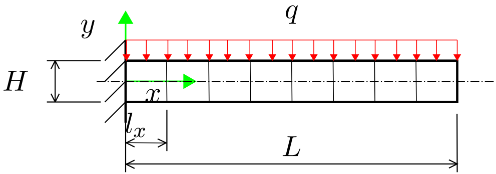
$$\delta W_{ext}^{gf} = \int_V \boldsymbol{b}^T \delta \boldsymbol{u}_h \, \mathrm{d}V = \underbrace{\int_V \boldsymbol{b}^T \boldsymbol{N}_m \, \mathrm{d}V}_{\boldsymbol{f}_{equiv}^{bf}} \delta \boldsymbol{u} \tag{18}$$

where $\boldsymbol{g}$ is the field of gravity. Equivalent nodal forces can be evaluated as follows:

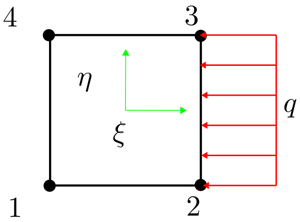
$$\boldsymbol{f}_{equiv}^{bf} = \rho \int_V \boldsymbol{g}^T \boldsymbol{N}_m \, \mathrm{d}V = \rho \int_{\Omega_R} \boldsymbol{g}^T \boldsymbol{N}_m \det \boldsymbol{J}_e \, \mathrm{d}\boldsymbol{\xi} \tag{19}$$

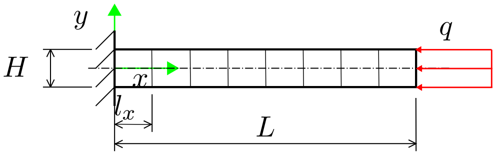
Example: Equivalent nodal forces from constant line load



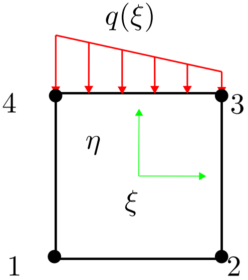


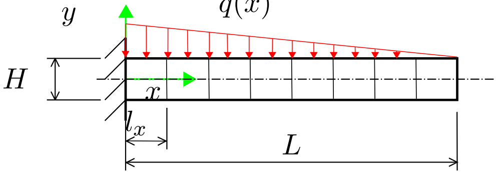
Bonus task: Equivalent nodal forces from constant line load in longitudinal direction





Bonus task: Equivalent nodal forces from linear line load





Bonus task: Equivalent nodal forces from gravity

Basics of FE-Analysis
BK10A6400
Lecture 10: 3D solid (continuum) elements

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3D solid elements

- Problems solving by "line" or "plane" elements based on rod, beam, or plane elements are special cases of three-dimensional solid (continuum) problems
- 3D solid elements are based on the full 3D elasticity, and no simplifications regarding geometry, strains, and stresses are needed.
- All six stresses (three normal and three shears) are included.
- Computationally demanding
- In the modeling of beam-, plate- and shell-type structures with solid elements, the element's shape (length/width relation) should become problematic (locking, ill-condition).

3D strains and stresses

Notations for strains and stresses in a vector form for a 3D case:

$$\boldsymbol{\varepsilon} = \begin{pmatrix} \varepsilon_{11} \\ \varepsilon_{22} \\ \varepsilon_{33} \\ 2\varepsilon_{12} \\ 2\varepsilon_{13} \\ 2\varepsilon_{23} \end{pmatrix} = \begin{pmatrix} \varepsilon_{xx} \\ \varepsilon_{yy} \\ \varepsilon_{zz} \\ \gamma_{xy} \\ \gamma_{xz} \\ \gamma_{yz} \end{pmatrix}$$

$$\boldsymbol{\sigma} = \begin{pmatrix} \sigma_{11} \\ \sigma_{22} \\ \sigma_{33} \\ \sigma_{12} \\ \sigma_{13} \\ \sigma_{23} \end{pmatrix} = \begin{pmatrix} \sigma_{xx} \\ \sigma_{yy} \\ \sigma_{zz} \\ \tau_{xy} \\ \tau_{xz} \\ \tau_{yz} \end{pmatrix}$$

Strain-stress relationship for linear elastic material (Hooke's law):

$$\boldsymbol{\sigma} = \boldsymbol{D}\boldsymbol{\varepsilon} \tag{20}$$

The elasticity matrix for isotropic material in terms of Lamé's parameters λ and G (often denoted by μ):

$$\boldsymbol{D} = \begin{pmatrix} \lambda + 2G & \lambda & \lambda & 0 & 0 & 0 \\ \lambda & \lambda + 2G & \lambda & 0 & 0 & 0 \\ \lambda & \lambda & \lambda + 2G & 0 & 0 & 0 \\ 0 & 0 & 0 & G & 0 & 0 \\ 0 & 0 & 0 & 0 & G & 0 \\ 0 & 0 & 0 & 0 & 0 & G \end{pmatrix} \tag{21}$$

$$\lambda = \frac{E\nu}{(1+\nu)(1-2\nu)} \tag{22}$$

$$G = \frac{E}{2(1+\nu)} \tag{23}$$

A linear relationship between strains and displacements (isoparametric description using bi-normalized local (natural) coordinates):

$$
\begin{aligned}
\varepsilon_{xx} &= \frac{\partial u_h}{\partial X_h} = \frac{\partial u_h}{\partial \xi}\left(\frac{\partial X_h}{\partial \xi}\right)^{-1} \\
\varepsilon_{yy} &= \frac{\partial v_h}{\partial Y_h} = \frac{\partial v_h}{\partial \eta}\left(\frac{\partial Y_h}{\partial \eta}\right)^{-1} \\
\varepsilon_{zz} &= \frac{\partial w_h}{\partial Z_h} = \frac{\partial w_h}{\partial \zeta}\left(\frac{\partial Z_h}{\partial \zeta}\right)^{-1} \\
\gamma_{xy} &= \frac{\partial u_h}{\partial Y_h} + \frac{\partial v_h}{\partial X_h} = \frac{\partial u_h}{\partial \eta}\left(\frac{\partial Y_h}{\partial \eta}\right)^{-1} + \frac{\partial v_h}{\partial \xi}\left(\frac{\partial X_h}{\partial \xi}\right)^{-1} \\
\gamma_{xz} &= \frac{\partial u_h}{\partial Z_h} + \frac{\partial w_h}{\partial X_h} = \frac{\partial u_h}{\partial \zeta}\left(\frac{\partial Z_h}{\partial \zeta}\right)^{-1} + \frac{\partial w_h}{\partial \xi}\left(\frac{\partial X_h}{\partial \xi}\right)^{-1} \\
\gamma_{yz} &= \frac{\partial v_h}{\partial Z_h} + \frac{\partial w_h}{\partial Y_h} = \frac{\partial v_h}{\partial \zeta}\left(\frac{\partial Z_h}{\partial \zeta}\right)^{-1} + \frac{\partial w_h}{\partial \eta}\left(\frac{\partial Z_h}{\partial \eta}\right)^{-1}
\end{aligned}
\tag{24}
$$

Displacement field $\boldsymbol{u}_h = \{u_h, v_h, w_h\}^T$ is interpolated as follows

$$\begin{cases} u_h(\xi,\eta,\zeta) := N_1 u_1 + N_2 u_2 + \ldots + N_n u_n \\ v_h(\xi,\eta,\zeta) := N_1 v_1 + N_2 v_2 + \ldots + N_n v_n \\ w_h(\xi,\eta,\zeta) := N_1 w_1 + N_2 w_2 + \ldots + N_n w_n \end{cases} \tag{25}$$

and same in a matrix form:

$$\underbrace{\begin{pmatrix} u_h \\ v_h \\ w_h \end{pmatrix}}_{\boldsymbol{u}_h} = \underbrace{\begin{pmatrix} N_1 & 0 & 0 & N_2 & 0 & 0 & \ldots & N_n & 0 & 0 \\ 0 & N_1 & 0 & 0 & N_2 & 0 & \ldots & 0 & N_n & 0 \\ 0 & 0 & N_1 & 0 & 0 & N_2 & \ldots & 0 & 0 & N_m \end{pmatrix}}_{\boldsymbol{N}_m} \underbrace{\begin{pmatrix} u_1 \\ v_1 \\ w_1 \\ u_2 \\ v_2 \\ w_2 \\ \vdots \\ u_n \\ v_n \\ w_n \end{pmatrix}}_{\boldsymbol{u}} \tag{26}$$

Position field

Position field $\boldsymbol{X}_h = \{X_h, Y_h, W_h\}^T$ is interpolated as follows;

$$\begin{cases} X_h(\xi,\eta,\zeta) := N_1 X_1 + N_2 X_2 + \ldots + N_n X_n \\ Y_h(\xi,\eta,\zeta) := N_1 Y_1 + N_2 Y_2 + \ldots + N_n Y_n \\ Z_h(\xi,\eta,\zeta) := N_1 Z_1 + N_2 Z_2 + \ldots + N_n Z_n \end{cases} \tag{27}$$

and same in a matrix form:

$$\underbrace{\begin{pmatrix} X_h \\ Y_h \\ Z_h \end{pmatrix}}_{\boldsymbol{X}_h} = \underbrace{\begin{pmatrix} N_1 & 0 & 0 & N_2 & 0 & 0 & \ldots & N_n & 0 & 0 \\ 0 & N_1 & 0 & 0 & N_2 & 0 & \ldots & 0 & N_n & 0 \\ 0 & 0 & N_1 & 0 & 0 & N_2 & \ldots & 0 & 0 & N_n \end{pmatrix}}_{\boldsymbol{N}_m} \underbrace{\begin{pmatrix} X_1 \\ Y_1 \\ Z_1 \\ X_2 \\ Y_2 \\ Z_2 \\ \vdots \\ X_n \\ Y_n \\ Z_n \end{pmatrix}}_{\boldsymbol{X}} \tag{28}$$

3D solid elements

Polynomial terms form Pascal's pyramid:

$$(\xi + \eta + \zeta)^n$$

8 node trilinear element

Basis within binormalized (natural) coordinates for eight-node trilinear element (24 dofs):

$$\{1, \xi, \eta, \zeta, \xi\eta, \xi\zeta, \eta\zeta, \xi\eta\zeta\} \tag{29}$$

Displacement field (polynomials form Pascal's pyramid):

$$\begin{cases} u_h(\xi,\eta,\zeta) := a_0 + a_1\xi + a_2\eta + a_3\zeta + a_4\xi\eta + a_5\xi\zeta + a_6\eta\zeta + a_7\xi\eta\zeta \\ v_h(\xi,\eta,\zeta) := b_0 + a_1\xi + b_2\eta + b_3\zeta + b_4\xi\eta + b_5\xi\zeta + b_6\eta\zeta + b_7\xi\eta\zeta \\ w_h(\xi,\eta,\zeta) := c_0 + a_1\xi + c_2\eta + c_3\zeta + c_4\xi\eta + c_5\xi\zeta + c_6\eta\zeta + c_7\xi\eta\zeta \end{cases}$$

8 node trilinear element

Displacement interpolation is the same for all displacements $\Rightarrow$ same shape functions. Eight unknowns, eight eqs:

$$\begin{cases} u_h(-1,-1,-1) := a_0 - a_1 - a_2 - a_3 + a_4 + a_5 + a_6 - a_7 = u_1 \\ u_h(1,-1,-1) := a_0 + a_1 - a_2 - a_3 - a_4 - a_5 + a_6 + a_7 = u_2 \\ u_h(1,1,-1) := a_0 + a_1 + a_2 - a_3 + a_4 - a_5 - a_6 - a_7 = u_3 \\ u_h(-1,1,-1) := a_0 - a_1 + a_2 - a_3 - a_4 - a_5 - a_6 + a_7 = u_4 \\ u_h(-1,-1,1) := a_0 - a_1 - a_2 + a_3 + a_4 - a_5 - a_6 + a_7 = u_5 \\ u_h(1,-1,1) := a_0 + a_1 - a_2 + a_3 - a_4 + a_5 - a_6\eta - a_7 = u_6 \\ u_h(1,1,1) := a_0 + a_1 + a_2 + a_3 + a_4 + a_5 + a_6 + a_7 = u_7 \\ u_h(-1,1,1) := a_0 - a_1 + a_2 + a_3 - a_4 - a_5 + a_6 - a_7 = u_8 \end{cases}.$$

In a matrix form:

$$\underbrace{\begin{pmatrix} 1 & -1 & -1 & -1 & 1 & 1 & 1 & -1 \\ 1 & 1 & -1 & -1 & -1 & -1 & 1 & 1 \\ 1 & 1 & 1 & -1 & 1 & -1 & -1 & -1 \\ 1 & -1 & 1 & -1 & -1 & -1 & -1 & 1 \\ 1 & -1 & -1 & 1 & 1 & -1 & -1 & 1 \\ 1 & 1 & -1 & 1 & 1 & -1 & -1 & 1 \\ 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ 1 & -1 & 1 & 1 & -1 & -1 & 1 & -1 \end{pmatrix}}_{\boldsymbol{A}} \underbrace{\begin{pmatrix} a_0 \\ a_1 \\ a_2 \\ a_3 \\ a_4 \\ a_5 \\ a_6 \\ a_7 \end{pmatrix}}_{\boldsymbol{a}} = \underbrace{\begin{pmatrix} u_1 \\ u_2 \\ u_3 \\ u_4 \\ u_5 \\ u_6 \\ u_7 \\ u_8 \end{pmatrix}}_{\boldsymbol{u}_u} . \tag{30}$$

8 node trilinear element

By solving a_i:s:

$$\boldsymbol{a} = \boldsymbol{A}^{-1}\boldsymbol{u}_u \tag{31}$$

and substituting them into:

$$u_h = \boldsymbol{a}^T\boldsymbol{p} = \boldsymbol{A}^{-T}\boldsymbol{u}_u^T\boldsymbol{p}^T = \underbrace{\boldsymbol{A}^{-T}\boldsymbol{p}}_{\boldsymbol{N}_{vec}}\boldsymbol{u}_u \tag{32}$$

where

$$\boldsymbol{p} = \{1, \xi, \eta, \zeta, \xi\eta, \xi\zeta, \eta\zeta, \xi\eta\zeta\} \tag{33}$$

and $\boldsymbol{N}_{vec}$ includes shape functions $N_1 \ldots N_8$

8 node trilinear brick (hexahedron) element

Shape functions are organized in a matrix form:

$$\boldsymbol{N} = \begin{pmatrix} N_1 & 0 & 0 & N_2 & 0 & 0 & \dots & N_8 & 0 & 0 \\ 0 & N_1 & 0 & 0 & N_2 & 0 & \dots & 0 & N_8 & 0 \\ 0 & 0 & N_1 & 0 & 0 & N_2 & \dots & 0 & 0 & N_8 \end{pmatrix} \tag{34}$$

Approximated displacement field:

$$\boldsymbol{u}_h = \boldsymbol{N}\boldsymbol{u} \tag{35}$$

Approximated position field:

$$\boldsymbol{X}_h = \boldsymbol{N}\boldsymbol{X} \tag{36}$$

Strain-stress relationship for linear elastic material (Hooke's law):

$$\boldsymbol{\sigma} = \boldsymbol{D}\boldsymbol{\varepsilon} \tag{37}$$

The elasticity matrix for isotropic material in terms of Lamé's parameters λ and G (often denoted by μ):

$$\boldsymbol{D} = \begin{pmatrix} \lambda + 2G & \lambda & \lambda & 0 & 0 & 0 \\ \lambda & \lambda + 2G & \lambda & 0 & 0 & 0 \\ \lambda & \lambda & \lambda + 2G & 0 & 0 & 0 \\ 0 & 0 & 0 & G & 0 & 0 \\ 0 & 0 & 0 & 0 & G & 0 \\ 0 & 0 & 0 & 0 & 0 & G \end{pmatrix} \tag{38}$$

$$\lambda = \frac{E\nu}{(1+\nu)(1-2\nu)} \tag{39}$$

$$G = \frac{E}{2(1+\nu)} \tag{40}$$

3D strains and stresses

A linear relationship between strains and displacements (isoparametric description using bi-normalized local (natural) coordinates):

$$
\begin{aligned}
\varepsilon_{xx} &= \frac{\partial u_h}{\partial X_h} = \frac{\partial u_h}{\partial \xi}\left(\frac{\partial X_h}{\partial \xi}\right)^{-1} \\
\varepsilon_{yy} &= \frac{\partial v_h}{\partial Y_h} = \frac{\partial v_h}{\partial \eta}\left(\frac{\partial Y_h}{\partial \eta}\right)^{-1} \\
\varepsilon_{zz} &= \frac{\partial w_h}{\partial Z_h} = \frac{\partial w_h}{\partial \zeta}\left(\frac{\partial Z_h}{\partial \zeta}\right)^{-1} \\
\gamma_{xy} &= \frac{\partial u_h}{\partial Y_h} + \frac{\partial v_h}{\partial X_h} = \frac{\partial u_h}{\partial \eta}\left(\frac{\partial Y_h}{\partial \eta}\right)^{-1} + \frac{\partial v_h}{\partial \xi}\left(\frac{\partial X_h}{\partial \xi}\right)^{-1} \\
\gamma_{xz} &= \frac{\partial u_h}{\partial Z_h} + \frac{\partial w_h}{\partial X_h} = \frac{\partial u_h}{\partial \zeta}\left(\frac{\partial Z_h}{\partial \zeta}\right)^{-1} + \frac{\partial w_h}{\partial \xi}\left(\frac{\partial X_h}{\partial \xi}\right)^{-1} \\
\gamma_{yz} &= \frac{\partial v_h}{\partial Z_h} + \frac{\partial w_h}{\partial Y_h} = \frac{\partial v_h}{\partial \zeta}\left(\frac{\partial Z_h}{\partial \zeta}\right)^{-1} + \frac{\partial w_h}{\partial \eta}\left(\frac{\partial Z_h}{\partial \eta}\right)^{-1}
\end{aligned}
\tag{41}
$$

Strain energy

$$W_{int} = \frac{1}{2}\boldsymbol{u}^T \underbrace{\int_x \int_y \int_z \boldsymbol{B}^T \boldsymbol{D} \boldsymbol{B} \, \mathrm{d}z \mathrm{d}y \mathrm{d}x}_{\overline{\boldsymbol{K}}} \, \boldsymbol{u}$$
$$= \frac{1}{2}\boldsymbol{u}^T \underbrace{\int_{-1}^{1} \int_{-1}^{1} \int_{-1}^{1} \boldsymbol{B}^T \boldsymbol{D} \boldsymbol{B} \, \det\left(\boldsymbol{J}_e\right) \mathrm{d}\xi \mathrm{d}\eta \mathrm{d}\zeta}_{\overline{\boldsymbol{K}}} \, \boldsymbol{u} \quad (42)$$

or more generally as (and no need to make integration at a component level):

$$W_{int} = \frac{1}{2}\int_V \boldsymbol{\sigma}\boldsymbol{\varepsilon} \, \mathrm{d}V = \frac{1}{2}\int_{-1}^{1} \int_{-1}^{1} \int_{-1}^{1} \boldsymbol{\sigma}\boldsymbol{\varepsilon} \det\left(\boldsymbol{J}_e\right) \mathrm{d}\xi \mathrm{d}\eta \mathrm{d}\zeta \quad (43)$$

$$\boldsymbol{f}_{int} = \frac{\partial W_{int}}{\partial \boldsymbol{u}} \quad ; \quad \overline{\boldsymbol{K}} = \frac{\partial \boldsymbol{f}_{int}}{\partial \boldsymbol{u}}$$

Numerical integration

How many Gaussian points are needed?
Interpolation: max polynomial order $\to 1$, $\boldsymbol{\varepsilon}^T \boldsymbol{\varepsilon} \to 1$
The degree of a polynomial of the integrand $p = 2$.

$$nip = \frac{(p+1)}{2} = 1.5 \tag{44}$$

so two integration points are needed. Reduced integration is needed for a shear part in the case of a trilinear solid element.